Unit 2 Progress Check: MCQ Part B

1. Which of the following code segments can be used to print only the even integers between 2 and 10, inclusive?

(A)

```
int num = 0;
while(num < 10)
{
  System.out.println(num);
  num += 2;
}
```

(B)

```
int num = 0;
while(num <= 10)
{
  System.out.println(num);
  num += 2;
}
```

(C)

```
int num = 0;
while(num < 10)
{
  num += 2;
  System.out.println(num);
}
```

(D)

```
int num = 0;
while(num <= 10)
{
  num += 2;
  System.out.println(num);
}
```

2. In the following code segment, x is an int variable with a positive value.

```
int temp = x;
while (temp > 0)
```

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```
{
  temp -= 2;
}
System.out.println(temp == 0);
```

Which of the following best describes the behavior of the code segment?

(A) It prints 0 if x is even and prints -1 otherwise.
(B) It prints -1 if x is even and prints 0 otherwise.
(C) It prints true if x is even and prints false otherwise.
(D) It prints false if x is even and prints true otherwise.

3. Consider the following code segment. Assume that the int variable input has been properly declared and initialized.

```
int answer = 1;
if (input != 0)
{
  int count = 1;

  while (count != input)
  {
    count++;
    answer *= count;
  }
}

System.out.println(answer);
```

Which of the following best describes the condition in which this code segment always results in integer overflow?

(A) When input == 0
(B) When input == 1
(C) When input > 0
(D) When input < 0

4. Consider the following code segment.

```
int a = 1;
String result = "";
while (a < 20)
{
  result += a;
  a += 5;
}
System.out.println(result);
```

What is printed as a result of executing this code segment?

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(A)
21

(B)
161116

(C)
161161

(D)
16111621

5. Consider the following code segment.

```
int sum = 0;
for (int j = 1; /* missing condition */; j += 2)
{
  sum += j;
}
```

Which of the following replacements for /* missing condition */ will cause an infinite loop?

(A)
j == 100

(B)
j != 100

(C)
j <= 100

(D)
j >= 100

6. Consider the following two code segments, which are intended to produce identical outputs.

Code Segment 1

```
for (int i = 0; i < 10; i++)
{
  System.out.println("counting " + i);
}
```

Code Segment 2

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```
int x = 0;
while (x < 10)
{
  x = x + 1;
  System.out.println("counting " + x);
}
```

Which of the following changes can be made to Code Segment 2 so that the outputs of the two code segments are identical as intended?

(A) Move the statement x = x + 1; so that it follows the print statement inside the loop body.
(B) Rename the variable x to be i so that it is the same variable name as in Code Segment 1.
(C) Remove the statement x = x + 1; because this statement is not in the loop body in Code Segment 1.
(D) Change the statement x = x + 1; to be x++; so that the increment of the loop control variable is the same in both code segments.

7. Consider the following code segment.

```
for (int m = 16; m > 0; m -= 2)
{
  if ((m % 3) == 1)
  {
    System.out.print(m + " ");
  }
}
```

What is printed as a result of executing this code segment?

(A)
15 12 9 6 3

(B)
12 6 0

(C)
16 14 12 10 8 6 4 2

(D)
16 10 4

8. In the following code segment, num has been assigned a positive int value. The following code segment is intended to print true if num is even and is intended to print false otherwise.

```
boolean isEven = true;
if (/* missing code */)
{
  isEven = false;
```

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```
}
System.out.println(isEven);
```

Which of the following can replace /* missing code */ so that this code segment works as intended?

(A)
num / 2 == 0

(B)
num / 2 != 0

(C)
num % 2 == 0

(D)
num % 2 != 0

9. Consider the following code segment.

```
int num = 245;
int temp = num;

while (temp > 0)
{
  System.out.print(temp % 10 + " ");
  temp /= 10;
}
```

What is printed as a result of executing this code segment?

(A)
2 4 5

(B)
4 5

(C)
5 4

(D)
5 4 2

10. In the following code segment, assume that low and high are properly declared and initialized int variables and that low < high. The code segment is intended to print the sum of the integers between low and high, inclusive, but does not always work as intended.

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```
int sum = 0;        // line 1
int j = low;        // line 2
while (sum <= high)  // line 3
{
  sum += j;         // line 5
  j++;              // line 6
}
System.out.println(sum);
```

Which of the following changes can be made so that this code segment works as intended?

(A) Changing line 1 to int sum = low;
(B) Changing line 2 to int j = 0;
(C) Changing line 3 to while (j <= high)
(D) Interchanging lines 5 and 6

11. In the following code segment, num has been assigned a positive int value.

```
int x = 0;
int temp = num;

while (temp > 0)
{
  x += temp % 10;
  temp /= 10;
}

System.out.println(x);
```

Which of the following best describes the value printed by this code segment?

(A) A count of the number of digits in num
(B) The average of the digits in num
(C) The sum of the digits in num
(D) Whether num is evenly divisible by 10

12. Consider the following code segment.

```
String str = "ABCDEF";
String result = "";
/* missing code */
System.out.println(result);
```

Which of the following can be used to replace `/* missing code */` so that this code segment prints the string `"EFCDAB"`?

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(A)

```
for (int j = str.length() - 1; j >= 0; j -= 1)
{
  result += str.substring(j, j + 1);
}
```

(B)

```
for (int j = str.length() - 1; j >= 0; j -= 2)
{
  result += str.substring(j, j + 2);
}
```

(C)

```
for (int j = str.length() - 2; j >= 0; j -= 1)
{
  result += str.substring(j, j + 2);
}
```

(D)

```
for (int j = str.length() - 2; j >= 0; j -= 2)
{
  result += str.substring(j, j + 2);
}
```

13. In the following code segment, str is a properly declared and initialized string. The code segment is intended to count the number of times that "a" appears in str.

```
int count = 0;
String temp = str;
int loc = temp.indexOf("a");

while (loc >= 0)
{
  /* missing code */
}
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

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(A)

```
count++;
temp = str.substring(loc + 1);
```

(B)

```
count++;
loc++;
temp = str.substring(loc);
```

(C)

```
count++;
temp = temp.substring(loc);
loc = temp.indexOf("a");
```

(D)

```
count++;
temp = temp.substring(loc + 1);
loc = temp.indexOf("a");
```

14. An acronym is formed by extracting the first letter of each word in a phrase of one or more words. For example, the acronym of the phrase "Laugh Out Loud" is "LOL", and the acronym of the phrase "If you know you know" is "Iykyk."

A phrase must meet the following conditions.

- The phrase has exactly one space between each word.
- The phrase does not have leading or trailing spaces.
- The final word in the phrase has at least two characters.

In the following code segment, the String variable phrase has been properly declared and initialized such that it meets these conditions. The code segment is intended to create the acronym of the String stored in phrase and store it in the variable text.

```
String temp = phrase;
String text = temp.substring(0, 1);
int i = temp.indexOf(" ");
while (i > 0)
{
  /* missing code */
}
```

Which of the following can be used to replace /* missing code */ so that the code segment works as intended?

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(A)

```
text += temp.substring(0, 1);
temp = temp.substring(i + 1);
i = temp.indexOf(" ");
```

(B)

```
temp = temp.substring(i + 1);
text += temp.substring(0, 1);
i = temp.indexOf(" ");
```

(C)

```
text += temp.substring(i, i + 1);
i = i + 1;
```

(D)

```
text += temp.substring(i, i + 1);
i = temp.indexOf(" ");
```

15. Consider the following code segment.

```
String str = "qrstu";
String result = "";
for (int j = 0; j < str.length(); j++)
{
  result += str.substring(0, j + 1);
}
System.out.println(result);
```

What, if anything, is printed as a result of executing this code segment?

(A)

rrsrstrstu

(B)

qqrqrsqrst

(C)

qqrqrsqrstqrstu

(D) Nothing is printed because an IndexOutOfBoundsException occurs.

16. Consider the following code segment.

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```
for (int j = 1; j <= 4; j++)       // Line 1
{
  for (int k = j + 1; k >= 1; k--)  // Line 3
  {
    System.out.print(j + " ");     // Line 5
  }
  System.out.println();
}
```

The code segment is intended to produce the following output, but does not work as intended.

```
1
2 2
3 3 3
4 4 4 4
```

Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 1 to `for (int j = 1; j < 4; j++)`
(B) Changing line 3 to `for (int k = 1; k <= j + 1; k++)`
(C) Changing line 3 to `for (int k = j; k >= 1; k--)`
(D) Changing line 5 to `System.out.print(k + " ");`

17. Consider the following code segment, in which `m` and `n` are properly declared and initialized `int` variables.

```
int result = 0;
for (int i = 0; i < n; i++)
{
  for (int j = m; j < n; j++)
  {
    result++;
  }
}
```

Which of the following best describes the conditions that must be true of `m` and `n` so that after this code segment executes, the value of `result` is greater than 0?

(A) $n > 0$ and $m > 0$
(B) $n > 0$ and $m > n$
(C) $n > 0$ and $n > m$
(D) $m > 0$ and $m > n$

18. Consider the following code segment.

```
int a = 1;
while (a <= 2)
```

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```
{
  int c = 1;
  while (/* missing condition */)
  {
    System.out.print("*");
    c++;
  }
  a++;
}
```

The code segment is intended to print "******".

Which of the following can be used to replace /* missing condition */ so that the code segment works as intended?

(A)
c > 2

(B)
c < 3

(C)
c <= 3

(D)
c >= 3

19. Consider the following code segment.

```
double sum = 0.0;
for (int outer = 1; outer <= 4; outer++)
{
   for (int inner = 0; inner <= 4; inner++)
   {
      sum += Math.random();
   }
}
```

How many times will Math.random() be called as a result of executing this code segment?

(A) 5
(B) 9
(C) 16
(D) 20

20. The following code segment appears in a method in the same class as the method doSomething().

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```
int i = 1;
while (i < 3)
{
   for (int j = 1; j <= 5; j++)
   {
      doSomething();
   }
   i++;
}
```

How many times will doSomething() be called as a result of executing this code segment?

(A) 5
(B) 8
(C) 10
(D) 15

21. Consider the following code segment.

```
for (int x = 1; x <= 3; x++)
{
   for (int y = 5; y > 1; y--)
   {
      System.out.print("*");
   }
}
```

Which of the following best describes the behavior of this code segment?

(A) The print statement is executed 7 times because the outer loop iterates 3 times and the inner loop iterates 4 times.
(B) The print statement is executed 8 times because the outer loop iterates 3 times and the inner loop iterates 5 times.
(C) The print statement is executed 12 times because the outer loop iterates 3 times and the inner loop iterates 4 times for each iteration of the outer loop.
(D) The print statement is executed 15 times because the outer loop iterates 3 times and the inner loop iterates 5 times for each iteration of the outer loop.